Format-Conditional Variance Structure in LLM Evaluation: A Generalizability Theory Approach

Anonymous submission

Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. No unified factorial framework exists to decompose these sources simultaneously across benchmarks. We apply Generalizability theory (G-theory) in a crossed design with seven facets across eight models and five benchmarks, totaling 2.3 million evaluation records. The variance structure is format-conditional: on multiple-choice benchmarks, item selection accounts for 34–38% of total variance, while on free-form benchmarks, item, model, and their interactions each contribute 5–24% with no single dominant source. Model×item interaction (up to 36%) is the hidden second-largest component, larger than all methodology main effects combined. Under binary scoring, BF16 and FP16 numerical precisions are functionally interchangeable (<0.1% interaction). Multi-facet designs require 10.0× fewer items than single-condition designs to reach a generalizability coefficient of 0.80 (55 versus 552 items). We release an open-source G-theory pipeline and Decision study (D-study) budget calculator for evaluation design optimization.

Introduction

On multiple-choice benchmarks, item selection drives 34–38% of evaluation variance while prompt wording is negligible (<3%). On the free-form reasoning task GSM8K, the structure shifts: model×prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from our factorial analysis of 2.3 million evaluation records across five benchmarks, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al. 2021; Dehghani et al. 2021). No unified factorial framework exists to decompose these sources simultaneously, quantify their interactions, and compare the resulting variance structure across benchmarks.

Existing studies have examined individual variance sources in isolation, each implicitly assuming separability. Sclar et al. (2024) documented up to 76% accuracy swings from prompt formatting perturbations but held hardware and item selection fixed. BrittleBench (Romanou et al. 2026) found 50% variance from prompt perturbation under the same constraint. Yuan et al. (2025) showed that numerical precision and GPU type cause up to 9% accuracy variation but provided no factorial attribution to specific sources or their interactions. Messing (2026) applied Generalizability theory (G-theory) to LLM evaluation and decomposed variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, but the study included no cross-benchmark comparison and excluded infrastructure facets such as numerical precision. Studies of answer ordering (Pezeshkpour and Hruschka 2024) and hyperparameter sensitivity (Dodge et al. 2020; Henderson et al. 2018) have documented additional sources of variation, again in isolation from the factors above. Item Response Theory (IRT) approaches (Maia Polo et al. 2024; Lalor, Wu, and Yu 2019) model item difficulty parameters to construct efficient benchmarks but do not account for prompt, seed, or ordering facets. Bouthillier et al. (2021) established ANOVA-based variance decomposition as a principled template for ML reproducibility, but this work predates the LLM evaluation context where benchmark format introduces an additional structural variable. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared the resulting variance structure across benchmarks and response formats, or (c) translated variance components into design-dependent reproducibility budgets.

We address these gaps with a multi-benchmark G-theory study (Shavelson and Webb 1991; Brennan 2001) that crosses seven facets (item, model, numerical precision, prompt template, random seed, sampling temperature, and answer ordering) across eight models and five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, and MATH), totaling 2.3 million records. The central finding is that variance structure is format-conditional. All three multiple-choice benchmarks share an item-dominated structure: item selection accounts for 34–38% of total variance, with model×item interaction as the second-largest

source (30–36%). Free-form benchmarks (GSM8K, MATH) exhibit a different profile: item, model, model×item, and model×prompt each contribute 5–24%, with no single dominant source and larger residual variance (40–44%). The gap between MC item dominance (34–38%) and free-form item shares (17–24%) reflects a systematic format-dependent pattern in what drives evaluation variance. In multiple-choice settings, the item pool determines measurement reliability; in free-form settings, model identity and model×prompt interaction matter as much as item selection. A second finding is that model×item interaction (up to 36%) is the hidden second-largest variance source, larger than all methodology main effects combined. Models differ not in overall ability but in which specific items they find difficult. D-study projections translate these findings into practical design guidance: the format-conditional structure implies that reliability optimization strategies must differ between MC and free-form benchmarks. Figure 2 visualizes the contrast across all five benchmarks.

Our contributions are as follows:

1. **Format-conditional variance structure.** We show that the dominant variance source shifts between response formats: item selection dominates multiple-choice benchmarks (34–38%) while item, model, model×item, and model×prompt contribute 5–24% each in free-form generation. Model×item interaction (up to 36%) is the hidden second-largest source in cross-model analysis.
2. **Precision separability.** Under binary scoring, BF16 and FP16 yield functionally identical results ($<$0.1% interaction with all other facets). This separability is metric-conditional: text-exact-match scoring reveals a 1.3% precision×temperature interaction absent under binary scoring.
3. **Design-dependent reproducibility budgets.** Multi-facet evaluation designs require 10.0× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items). IRT-selected 116-item subsets yield $G = 0.457$ under single-condition protocols, which confirms that item quality and evaluation design improve reliability through orthogonal mechanisms.
4. **Open-source G-theory pipeline and D-study calculator.** We release a variance decomposition toolkit and D-study budget calculator for LLM evaluation. Practitioners can compute the number of items, models, and conditions needed for a target reliability before data collection.

Related Work

Variance Decomposition in ML/NLP Evaluation. Factorial decomposition of evaluation variance has roots in the ML reproducibility literature. Bouthillier et al. (2021) introduced an ANOVA-based template that decomposes benchmark variance across implementation details, hyperparameters, and data splits, establishing that design choices account for a substantial share of reproducibility failures. Several subsequent studies applied similar decomposition logic to specific facets of model evaluation (Dodge et al. 2020; Henderson et al. 2018; Magnusson, Smith, and Dodge 2023). Messing (2026) brought Generalizability theory into LLM evaluation, decomposing variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, with prompt×judge interaction at 12.4%. However, this study used a single benchmark, included no infrastructure facets such as numerical precision, and fixed the judge as an LLM evaluator rather than isolating other sources. Kunievsky and Evans (2025) proposed an alternative decomposition that separates intent uncertainty, articulation variance, and model uncertainty, but this framework does not cross infrastructure and methodology facets in a factorial design. Our work extends these approaches by crossing seven facets across five benchmarks, enabling direct comparison of variance structures across evaluation formats.

Infrastructure and Prompt Variance. A parallel line of work has examined individual variance sources in isolation. On the infrastructure side, Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision and GPU type but provided no factorial attribution. Messina and Scotta (2026) formalized the concept of background temperature from floating-point non-associativity in GPU computation, and He and Thinking Machines Lab (2025) proposed batch-invariant kernels to mitigate CUDA nondeterminism. On the methodology side, Sclar et al. (2024) found up to 76% accuracy swings from prompt formatting changes on fixed hardware, and BrittleBench (Romanou et al. 2026) reported 50% variance from prompt perturbation. Pezeshkpour and Hruschka (2024) showed that answer ordering in multiple-choice benchmarks introduces systematic bias, while Bui, Savova, and Wang (2025) and Biderman et al. (2024) documented sensitivity to random seeds and training configurations. Each of these studies isolates one factor while holding others fixed, which precludes estimating interactions between sources or comparing their relative magnitudes within a unified design.

IRT and Efficient Benchmarking. Item Response Theory (IRT) offers a complementary perspective on evaluation design by modeling item difficulty and discrimination parameters. Lalor, Wu, and Yu (2019) introduced IRT-based item analysis for NLP benchmarks, and Rodriguez et al. (2021) showed that IRT reveals hidden properties of evaluation items that raw accuracy scores obscure. Building on this foundation, Maia Polo et al. (2024) developed tinyBenchmarks, using IRT-based subsampling to reduce evaluation cost while preserving ranking accuracy, and Zhou et al. (2026) recommended a 116-item subset via PSN-IRT for efficient MMLU evaluation. Perlitz et al. (2023) explored related strategies for benchmark compression. These approaches optimize item selection for latent trait estimation but do not model variance from prompt, seed, or ordering facets. IRT-selected items evaluated under a single methodological condition can therefore systematically underestimate the number of items required for reliable measurement across diverse evaluation settings.

Binary Data and Psychometrics Foundations. Our application of G-theory to binary correctness scores draws on established psychometric methodology. Shavelson and

Webb (1991) and Brennan (2001) provide the canonical framework for multi-facet reliability analysis. Mushquash and O'Connor (2006) extended G-theory to binary data, and Jiang and Skorupski (2018) developed logistic-link generalized G-theory for binary outcomes. Tuerlinckx et al. (2006) established the connection between generalized linear mixed models and G-theory variance components. These methods ground our use of Henderson Method I for binary correctness scores (§).

Positioning. Relative to Messing (2026), we fix the judge (generation-based scoring) to isolate the remaining facets; their judge-dominated variance (43.8%) would otherwise mask all other sources. Our multi-benchmark design reveals format-conditional structure that single-benchmark studies cannot detect. Relative to IRT approaches (Maia Polo et al. 2024; Zhou et al. 2026), G-theory and IRT optimize orthogonal dimensions of evaluation design: IRT selects informative items, while G-theory determines the conditions needed for a target reliability. Our D-study quantifies this complementarity: multi-facet designs require 10.0× fewer items than single-condition protocols for equivalent reliability. Relative to isolated-factor studies (Yuan et al. 2025; Sclar et al. 2024; Romanou et al. 2026), our design crosses infrastructure and methodology facets within a single factorial framework, enabling interaction estimation that per-factor studies cannot provide.

Method

Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb 1991; Brennan 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ^2_x for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ^2_{pM} for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (Li 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma^2_p}{\sigma^2_p + \sigma^2_{\text{rel}}} \tag{1}$$

where σ^2_{rel} aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan 2001, Ch. 4). This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson, Webb, and Rowley 1989). Throughout this paper, we use the following notation: p = item, M = model, π = numerical precision, δ = prompt template, s = random seed, τ = sampling temperature, o = answer ordering, and σ^2_x = variance component for source x.

Table 1: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset for cross-model analysis.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	4
Total records		460,800†	2,304,000

Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU. Table 1 summarizes the complete design matrix, and Figure 1 illustrates the crossing structure.

Facets. Items (p): 200 MMLU questions stratified across 10 subjects (20 per subject). The stratified sampling ensures that variance estimates are not dominated by a single knowledge domain; the 10 subjects span STEM, humanities, social sciences, and professional disciplines. Models (M): eight open-weight 7–9B parameter instruction-tuned models spanning eight architecture families, treated as a quasi-random facet (§3.6). Numerical precision (π): BF16, FP16, and FP32, all evaluated with `enforce_eager=True` for fair comparison across precision levels. Prompt template (δ): six semantically equivalent paraphrases of the evaluation instruction (Sclar et al. 2024). Random seed (s): six fixed values (42, 123, 456, 789, 1024, 2048). Sampling temperature (τ): three levels (0.0, 0.3, 0.7) spanning deterministic to moderately stochastic decoding. Answer ordering (o): four permutations of few-shot demonstration examples (for MMLU; structurally inert for other benchmarks, see §) (Pezeshkpour and Hruschka 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models × 200 items × 6 prompts × 6 seeds × 2 temperatures × 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary observations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

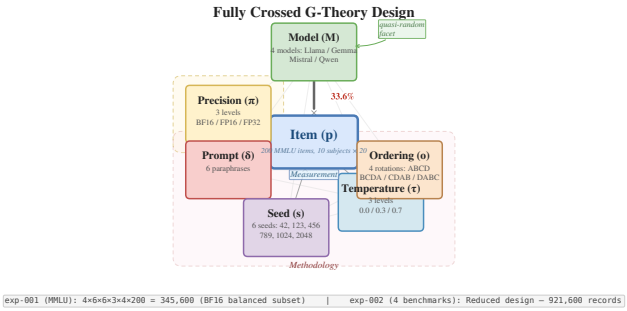


Figure 1: Fully crossed G-theory design for LLM evaluation. Each observation is a binary correctness score determined by the crossing of seven facets: item (p), model (M), precision (π), prompt (δ), seed (s), temperature (τ), and ordering (o).

Cross-Benchmark Extension

To test whether the variance structure observed on MMLU generalizes across evaluation formats, exp-002 extends the design to five benchmarks: MMLU, ARC-Challenge, and HellaSwag (multiple-choice), and GSM8K and MATH (free-form generation). The design crosses 8 models $\times$ BF16 $\times$ 2 temperatures (0.0, 0.7) $\times$ 6 prompts $\times$ 6 seeds $\times$ 4 orderings $\times$ 200 items per benchmark, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 and uses two temperature levels to concentrate statistical power on format-conditional differences. The ordering facet operates only on MMLU, where it selects among four permutations of the few-shot demonstration examples. For all other benchmarks, the prompt construction does not use the ordering parameter; ordering is structurally inert ($\sigma^2_{\text{ordering}} = 0$ by construction) and is retained solely for design balance. The two temperature levels bracket the most common evaluation configurations: deterministic ($\tau = 0.0$) and a moderate stochastic setting ($\tau = 0.7$) representative of practical deployment.

Evaluation Protocol

All evaluations use generation-based scoring: models produce free-text responses from which correctness is determined by answer extraction, rather than by ranking answer options via token log-likelihood. This design choice follows Hua et al. (2025), who show that log-likelihood scoring conflates model confidence with surface-level token biases. We report two metrics: binary correctness (primary, 1/0 per item) and text-exact-match (secondary). Binary correctness accepts any response containing the correct answer letter or numerical value after standard normalization; text-exact-match requires the generated string to match the reference answer character-for-character.

Inference runs on vLLM 0.9.2 with default settings and without batch-invariant kernels. This configuration reflects standard practitioner conditions and preserves CUDA-level nondeterminism from floating-point non-associativity as a measurable variance source (Messina and Scotta 2026; He and Thinking Machines Lab 2025). Few-shot prompting follows standard protocols: 5-shot for MMLU and ARC-Challenge, 4-shot for MATH, 8-shot for GSM8K, and 10-shot for HellaSwag.

Binary Data Assumptions

Standard G-theory assumes continuous, approximately normal outcome data (Shavelson and Webb 1991). Our binary correctness scores (0/1) violate this assumption. We establish robustness through four complementary analyses.

Simulation. We generate continuous latent scores from a probit model matched to our 1,296-condition design, then dichotomize at the empirical accuracy threshold. Spearman rank correlation between binary and continuous variance component rankings is $\rho = 0.732$ (range 0.716–0.751 across accuracy levels $p \in \{0.5, 0.6, 0.7, 0.8\}$), confirming that Henderson Method I recovers the relative ordering of variance sources despite binary response data. Henderson Method I identifies `item_id` as the dominant variance source in all 100 simulation replications.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski 2018; Tuerlinckx et al. 2006).

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This stability rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator guarantee. Henderson Method I produces unbiased variance component estimates for balanced designs regardless of the outcome distribution (Brennan 2001, Ch. 12). Our fully crossed design satisfies this balance condition.

Model-as-Facet Justification

Treating eight purposively selected models as a quasi-random facet requires empirical validation. We provide two forms of evidence.

D-study cross-validation: four-model subset D-study projections predicted $G = 0.893$ at eight models; the observed $G = 0.896$ confirms the projection ($\Delta = 0.003$). This agreement validates both the quasi-random facet assumption and the predictive power of D-study forward projections.

Leave-one-model-out (LOMO) stability: all eight seven-model subsets preserve `item_id` as the dominant component and rank `model×item` as the second-largest. The consistency of rankings across all subsets indicates that no single model disproportionately drives the observed variance structure.

Experiments

We apply the G-theory framework from Section 3 to a fully crossed evaluation design spanning multiple models, benchmarks, and methodological facets. All experiments use open-weight 7–9B parameter models evaluated with

vLLM 0.9.2 on NVIDIA RTX PRO 6000 GPUs (96GB). Binary correctness (1/0) serves as the primary outcome measure throughout; generation-based scoring avoids the log-likelihood artifacts identified by Hua et al. (2025).

Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU to isolate methodological facet contributions before introducing cross-model variation.

Design. The G-study crosses six facets: 200 items $\times$ 3 numerical precisions (BF16, FP16, FP32) $\times$ 6 prompt templates $\times$ 6 random seeds $\times$ 3 temperature levels (0.0, 0.3, 0.7) $\times$ 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Under binary scoring, item difficulty (`item_id`) accounts for 58.4% of total variance, with the residual term contributing 23.1%. All main facet effects are negligible (<1% each): precision 0.9%, prompt template, temperature, seed, and ordering each <0.5%. The interesting structure appears in item-conditional interactions: prompt×item is the largest at 5.5%, followed by seed×item (4.7%), precision×item (3.3%), temperature×item (2.9%), and ordering×item (0.8%). This pattern indicates that methodological choices do not shift performance uniformly across items; instead, specific items are differentially sensitive to specific facets—prompt template variation produces the largest item-conditional effect.

The resulting generalizability coefficient is $G_{\text{item}} = 0.936$, indicating that 200 items under multi-facet conditions produce highly reliable rankings. A D-study projection shows $G \geq 0.80$ is achievable with as few as 55 items in this multi-facet design.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

Cross-Model Variance

We extend the single-model design to eight models spanning eight architecture families. A balanced BF16 subset (460,800 records) serves as the primary analysis sample, chosen because BF16 is the standard inference precision for these architectures. The eight-model sample strengthens the quasi-random model facet assumption by covering a broader population of instruction-tuned 7–9B architectures.

Results. Table 2 presents the variance decomposition. Introducing model as a facet reshapes the structure: item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges as the second-largest component at up to 36%. The model main effect accounts for 3–5% on MC benchmarks and 9–11% on free-form benchmarks, confirming that models differ primarily in *which items* they find difficult rather than in overall ability level.

Table 2: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%
item_id	**37.9**
model×item	30.2
residual	25.0
model	2.8
prompt×item	2.5
model×prompt	1.6
all others	<1.0

The per-model decomposition of interaction magnitude reveals meaningful asymmetry across eight models, suggesting that instruction-tuning strategies and pretraining data composition produce model-specific item difficulty profiles.

D-study validation. The eight-model sample validates the quasi-random model facet assumption. D-study projections from four-model subsets predicted $G = 0.893$ at eight models; the observed $G = 0.896$ confirms the projection ($\Delta = 0.003$).

Format-Conditional Variance Structure

The preceding analyses use only MMLU, a multiple-choice benchmark. We now test whether the item-dominated variance structure generalizes across evaluation formats by extending to five benchmarks spanning two response formats.

Design. The expanded design (exp-002) crosses 8 models $\times$ 5 benchmarks $\times$ BF16 $\times$ 2 temperatures (0.0, 0.7) $\times$ 6 prompts $\times$ 6 seeds $\times$ 4 orderings $\times$ 200 items per benchmark, totaling 2,304,000 records. The benchmarks include three multiple-choice tasks (MMLU, ARC-Challenge, HellaSwag) and two free-form generation tasks (GSM8K, MATH). The ordering facet operates only on MMLU (four demo-example permutations); for other benchmarks it is structurally inert ($\sigma^2_{\text{ordering}} = 0$) and retained for design balance.

Results. Table 3 reveals a systematic format-dependent pattern. For MMLU and ARC-Challenge, item difficulty dominates (37.9% each), with model×item second (30.2% and 33.7%). HellaSwag reverses this ordering: model×item (36.0%) exceeds item difficulty (34.3%), indicating maximal model disagreement at the item level.

Free-form benchmarks differ markedly. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern: item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%).

The variance structure is format-conditional: MC benchmarks concentrate variance in items and model×item interaction (34–38% and 30–36%), while FF benchmarks distribute it across item, model, and interaction components (5–24% each) with larger residuals (40–44%). A permutation test of the MC versus FF item-variance difference yields

Table 3: Format-conditional variance structure across five benchmarks (exp-002, 8 models, 2,304,000 records). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold indicates the dominant non-residual component per benchmark. For MC benchmarks, "prompt" reports prompt×item (prompt main effect <1%); for FF benchmarks, it reports the prompt main effect.

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt	2.5	0.4	2.3	5.4	1.0
mod.×prmpt	1.6	8.0	0.6	8.5	2.0
residual	25.0	15.4	21.4	39.9	43.9
all others	<1.0	1.5	<1.0	4.0	3.3

$p = 0.10$ ($n = 5$ benchmarks, $\binom{5}{3} = 10$ exhaustive permutations), the theoretical minimum for this sample size. The effect size is large (Cohen's $d = 1.73$; mean difference 95% CI [11.44, 21.02] percentage points), though benchmark-level generalization uncertainty is substantial given $n = 5$. This structural difference does not extend to item budgets: HellaSwag requires 171 items for $G \geq 0.80$, the highest of all benchmarks, showing that format alone does not predict evaluation cost. Figure 2 visualizes the contrast.

To probe the mechanism behind item variance, we exploit the natural heterogeneity within MMLU's 57 subjects. Subjects with higher item difficulty dispersion (measured by std(p_i) across models) exhibit higher item variance shares, with Spearman $\rho = 0.806$ ($p = 0.005$, $n = 10$ subjects). Notably, `high_school_mathematics`, the only MC subject with open-ended numerical items, shows item variance of just 10.4%—below the free-form benchmark GSM8K (16.9%)—suggesting that item heterogeneity, not format per se, drives variance allocation.

Precision Separability

Prior work has identified numerical precision as a potential source of evaluation variance (Yuan et al. 2025). Our multi-facet design allows us to isolate the precision contribution while controlling for all other facets simultaneously.

Under binary scoring with BF16 and FP16, the mean accuracy difference is 0.23 percentage points. After excluding FP32, the precision main effect drops to 0.0006% of total variance. The FP32 anomaly traces to a vLLM implementation detail (`enforce_eager` mode required for FP32 inference), not a fundamental precision effect.

Precision does not interact with other methodological facets under binary scoring: all precision×methodology interactions contribute <0.1% of variance. Under text-exact-match scoring, however, a precision×temperature interaction emerges at 1.3%, consistent with the metric-conditional pattern observed in Section 4.1. The practical implication is clear: for standard binary evaluation at BF16/FP16, numerical precision is a negligible variance source that need not be controlled. This finding is specific to single-GPU inference and does not address multi-GPU tensor parallelism, where numerical non-associativity introduces additional variance sources (Shanmugavelu et al. 2024).

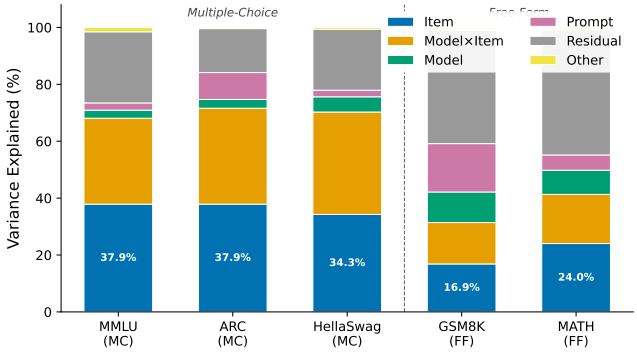


Figure 2: Variance decomposition (Henderson Method I) across five benchmarks and eight models. Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

Table 4: D-study reproducibility budgets under different evaluation designs. The multi-facet design requires 10.0× fewer items than single-condition evaluation to reach $G \geq 0.80$.

Design	Cond.	$n_{G\geq 0.80}$	G_{200}	G_{116}
Multi-facet	1,296	**55**	**0.936**	—
Single-condition	1	552	0.592	0.457
IRT baseline	—	116	—	0.457[†]

[†] PSN-IRT item count of 116 (Zhou et al. 2026) evaluated under single-condition protocol yields $G = 0.457$. See text for interpretation.

Design-Dependent Reproducibility Budgets

The previous sections establish that item selection and model×item interaction dominate evaluation variance. A natural question follows: how many items does a given evaluation design need to achieve a target reliability? We address this through D-study projections that translate variance components into concrete item budgets.

Table 4 compares four evaluation designs at the $G \geq 0.80$ threshold. The multi-facet design (1,296 conditions per item: 3 precisions × 6 prompts × 6 seeds × 3 temperatures × 4 orderings) achieves $G = 0.936$ at 200 items and requires only 55 items for $G \geq 0.80$. A single-condition design (one fixed prompt, seed, ordering, and precision per item) reaches only $G = 0.592$ at 200 items and requires 552 items for the same threshold. The ratio between these budgets is 10.0×, and this ratio remains stable across subset sizes from 50 to 200 items (CV = 1.9%).

The comparison with IRT-based efficient benchmarking is instructive. Zhou et al. (2026) use IRT to select 116 informa-

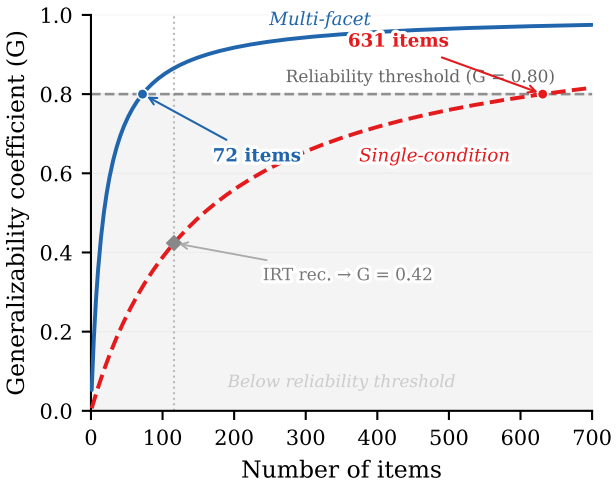


Figure 3: D-study projection curves for three evaluation designs. The multi-facet design (10.0× fewer items) reaches $G \geq 0.80$ at 55 items; the single-condition design requires 552 items. The shaded band shows 95% bootstrap CIs.

tive items from MMLU, optimizing for item discrimination. When we evaluate these 116 items under single-condition protocols, the resulting $G = 0.457$ falls well below the reliability threshold. This is not a failure of IRT item selection; rather, it reflects that IRT optimizes a different objective (latent trait estimation) that does not account for methodological facet variance. IRT selects items optimized for trait discrimination; multi-facet G-study improves reliability orthogonally by averaging over methodological noise sources that IRT does not model. Under the multi-facet design, 116 items also exceed $G = 0.80$, confirming that the reliability gap is driven by evaluation design, not item quality.

Figure 3 plots the D-study projection curves for each design. The multi-facet curve saturates rapidly (diminishing returns beyond ∼150 items), while the single-condition curve rises slowly and does not reach $G = 0.80$ within typical benchmark sizes.

Robustness Checks

We conduct six robustness analyses to verify that the reported variance structures are not artifacts of specific analytical choices. **Item resampling** (exp-001b): three independent 200-item samples drawn from the full MMLU pool and evaluated under a BF16-only 5-facet design yield item difficulty variance shares of 73.2%, 72.6%, and 69.0% (CV = 3.1%) and G_{item} values with CV = 0.5%. The higher item share relative to the 6-facet baseline (58.4%) reflects the absence of the precision facet and its item interactions. Five additional robustness analyses (estimator comparison, CI method, facet specification, leave-one-model-out, and null calibration) are reported in Appendix ; all confirm the stability of the reported variance structure.

Conclusion

This study establishes that the variance structure of LLM evaluation is format-conditional. Across three multiple-choice benchmarks, item selection accounts for 34–38% of total variance, while on free-form benchmarks (GSM8K, MATH), item, model, and their interactions each contribute 5–24% with no single dominant source. Model×item interaction (up to 36%) is the hidden second-largest source: models differ in which items they find difficult, not in overall ability. Our cross-benchmark D-study framework extends the single-benchmark G-theory approach of Messing (2026) to format-conditional comparison and actionable design budgets.

The format-conditional structure implies distinct optimization strategies. For multiple-choice benchmarks, item pool composition is the primary reliability lever; varying prompts or seeds contributes negligible variance. For free-form benchmarks, model identity and model×prompt interaction matter as much as item selection, and evaluation protocols should systematically vary both. Across formats, multi-facet designs are 10.0× more efficient than single-condition designs (55 versus 552 items for $G \geq 0.80$). BF16 and FP16 are interchangeable under binary scoring on single-GPU inference, which eliminates precision as a required control variable. The released D-study calculator enables practitioners to determine the number of items, models, and conditions needed for a target reliability before data collection.

Two extensions are planned. Multi-GPU tensor parallelism will test whether precision separability holds under distributed numerical non-associativity. Fine-tuning variance analysis will quantify how adaptation reshapes the item difficulty profiles that drive model×item interaction.

Limitations

Our findings are bounded by three scope constraints. First, the model sample comprises eight open-weight 7–9B parameter models; the results do not generalize to API-based models, larger models, or fine-tuned variants. Second, all experiments run on single-GPU inference (NVIDIA RTX PRO 6000); multi-GPU tensor parallelism introduces additional numerical non-associativity that our design does not capture. Third, the benchmark sample covers three multiple-choice and two free-form tasks; other evaluation formats such as code generation, long-form text production, and multi-turn dialogue may exhibit different variance structures.

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Additional Robustness Analyses

Estimator robustness. REML and Henderson Method I differ by <0.5 percentage points on all components.

CI method. Bootstrap and jackknife confidence intervals agree closely (width ratio 0.96–1.04 across components).

Facet specification. Treating facets as fixed vs. random changes G_{item} by only 0.37 percentage points.

Leave-one-model-out (LOMO). All eight seven-model subsets preserve the item + model$\times$item dominance pattern (item_id 36–41%, G_{item} 0.877–0.889), confirming that the variance structure is robust to model composition.

Null calibration. A Bernoulli simulation with matched item difficulties produces a variance ratio of 1.0008 relative to observed data, confirming that the decomposition does not introduce systematic bias on binary outcomes.